



CS & IT ENGINEERING

Discrete Mathematics



Mathematical logic

Lecture No.- 01



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Recap of Previous Lecture



Topic

Extra Topics in Graph Theory



Topics to be Covered

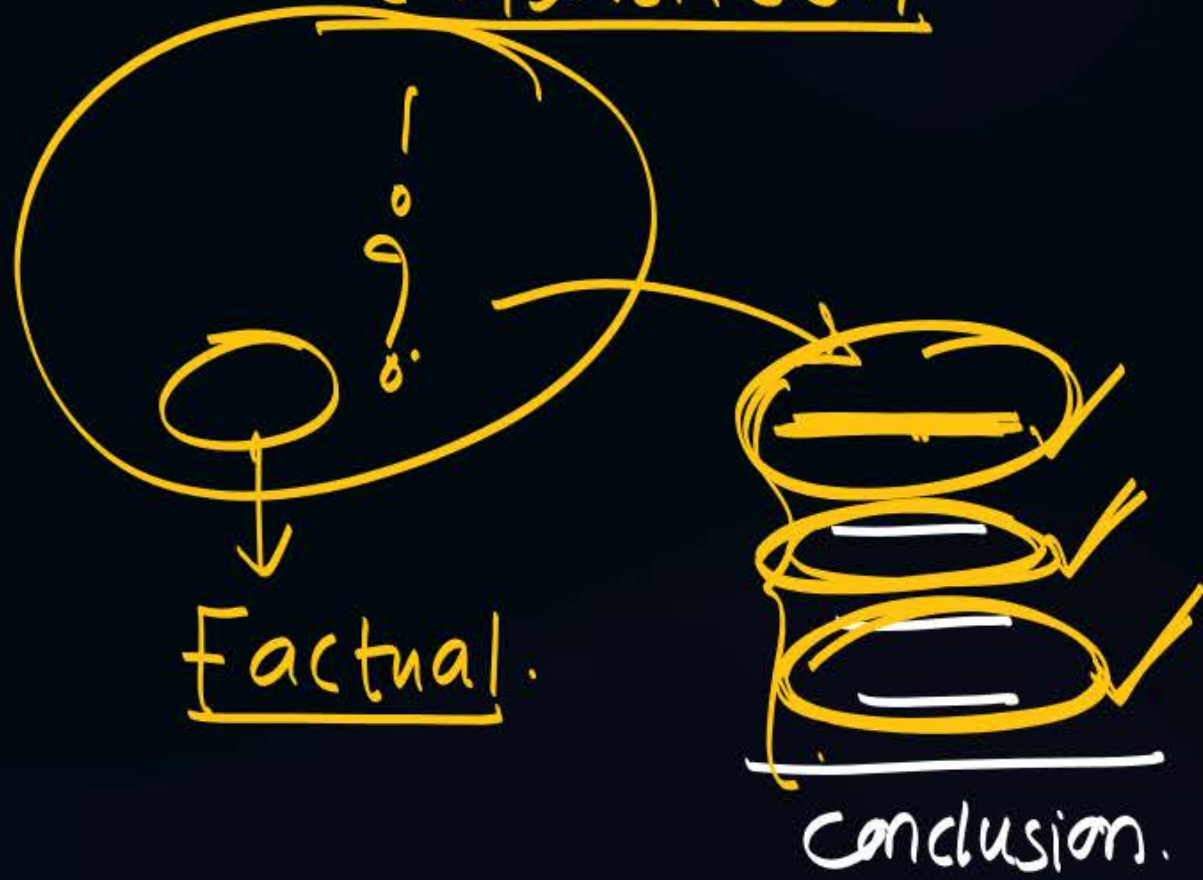


Topic

Logic Part-01



English stm:



→ Thms. → Algo → PL
↓
S/W



propositional stmt. $\begin{cases} \rightarrow \text{Yes} \\ \rightarrow \text{No} \end{cases}$

Simple.

can not break.

$5 > 3 (T)$

Compound:

breakable.

$5 \geq 3$

$5 > 3 \text{ OR } 5 = 3$

connective:

(breakable point)

compound:

S_1

S_2

Connectives

$\rightarrow$ conjunction.

$\rightarrow$ Disjunction.

$\rightarrow$ Conditional.

$\rightarrow$ Biconditional.

$S_1 \quad S_2 \quad | \quad (C)$

conjunction: ($\wedge$) (AND/but)

<u>P</u>	<u>Q</u>	<u>$P \wedge Q$</u>
T	F	F
F	T	F
F	F	F
T	T	T

compound:

$$\underline{F} \wedge \underline{\quad} \equiv F$$

$$\underline{F} \wedge \underline{\quad} \equiv F$$

AND hates false

OR $\rightarrow$ Inclusive OR ^($\vee$) (one, other, both)
 $\rightarrow$ Exclusive OR ^($\oplus$) (one or other)
 (XOR)

$$\underline{T} \vee \underline{\quad} \equiv T$$

OR Loves True.

p	q	$p \vee q$
T	F	T
F	T	T
T	T	T
F	F	F

conditional stmt ($\rightarrow$)

if p then q .
 p implies q .
 if p , q .
 q when p .
 q whenever p
 q unless $\neg p$
 p only if q
 q if p .

$p \rightarrow q$

Rosen

* if you clear the GATE then she will marry you.

P	Q	$P \rightarrow Q$
$\rightarrow$ clear(T)	marry(T)	T
clear(T)	marry(F)	(F)
clear(F)	marry(T)	(T)
clear(F)	marry(F)	T

$$\left\{ \begin{array}{l} T \rightarrow T \equiv T \\ T \rightarrow F \equiv F \\ F \rightarrow T \equiv T \\ F \rightarrow F \equiv T \end{array} \right.$$

Thm:

if x is even then x^2 is also even.

integer

○
↑
~~~~~

$x$  is even <sup>(T)</sup>  
 $x=2$

$x^2$  even <sup>T</sup>  
 $x^2=4$

$T \rightarrow T \equiv T$

○  
↑  
~~~~~

x even ^(T)

x^2 ~~even~~ ^(F)

$T \rightarrow F \equiv \textcircled{F} \rightarrow$ Thm is wrong.

○
↑
~~~~~

~~$x$  even~~ <sup>(F)</sup>  
 $x=3$

$x^2$  even

$\textcircled{F} \rightarrow T \equiv T$   
 $\textcircled{F} \rightarrow F \equiv F$

○  
↑

$x$  even

$x^2$  even



$$P \rightarrow Q$$

if clear GATE then she marry you.  
P Q

## Inverse.

$$\neg p \rightarrow \neg q$$

if you will not clear  
GATE then she will not  
marry.

Converse.

$$q \rightarrow p$$

if she marry you  
then you clear  
the GATE.

## Contrapositive:

79 → 7P.

i.f she not married  
then you have not  
cleared.

$$P \rightarrow Q$$

is same as

$$\gamma q \Rightarrow \gamma p$$

→ logical equivalent

$$p \rightarrow q \equiv \neg q \rightarrow \neg p.$$

power the LED-b'

$$p \rightarrow q.$$

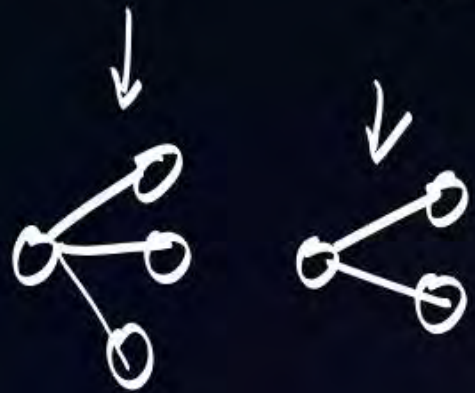
$$p.m \rightarrow \text{even}$$

$$\neg p \rightarrow \neg q.$$

$$q \rightarrow p$$

$$\neg q \rightarrow \neg p.$$

if p.m not then no. of  
exist vertices  
will not be  
even



if no. of vertices are not even then p.m will not  
exist.



| P | Q | $\neg P$ | $\neg Q$ | A<br>$P \rightarrow Q$ | B<br>$\neg Q \rightarrow \neg P$ |
|---|---|----------|----------|------------------------|----------------------------------|
| T | T | F        | F        | T                      | T                                |
| T | F | F        | T        | F                      | F                                |
| F | T | T        | F        | T                      | T                                |
| F | F | T        | T        | T                      | T                                |

logical equivalent  
 $A \equiv B$   
 \* column of A = column of B.  
 \* A, B are having same behaviour.

$$\frac{P \rightarrow Q}{\neg Q} = \frac{\neg Q \rightarrow \neg P}{\neg P}$$



## 2 mins Summary

Topic

One

Topic

Two

Topic

Three

Topic

Four

Topic

Five

# Logic Part-01







**THANK - YOU**